

Homogeneous quasinormal modes of a maximum-likelihood lattice-calibrated holographic plasma

ABSTRACT: We compute all three homogeneous quasinormal sectors of the maximum-likelihood realization of a lattice-calibrated Einstein–Maxwell–dilaton model. Reference-compatible continuation and an independent cusp condition locate the critical endpoint at approximately 103.755 MeV and 593.323 MeV and resolve the coexistence and spinodal lines. At this background, source-factored Chebyshev collocation and complex shooting independently validate the leading quintuplet and singlet poles and two triplet poles. Their least-damped dimensionless rates are 0.8669, 0.1032, and 0.2561, respectively. Along zero baryon chemical potential and a fixed chemical-potential-to-temperature ratio of two, the triplet rate decreases nonmonotonically to 0.0037 while the pole remains oscillatory. Point-wise shooting and eigenfunction overlaps above 0.9985 confirm both trajectories. Repeating the calculation with the earlier 2018 EMD parameters reproduces nine published relaxation times within 0.012 fm/c. The analysis is restricted to the maximum-likelihood realization and to homogeneous response; it neither represents posterior uncertainty nor infers diffusion, sound, or spinodal growth.

KEYWORDS: Gauge-Gravity Correspondence, Non-Equilibrium Field Theory, Finite Temperature or Finite Density, Quark-Gluon Plasma

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1 Introduction

Real-time response at nonzero baryon density is a central open problem in the theory of the quark–gluon plasma. Lattice QCD tightly constrains equilibrium physics around vanishing baryon density, but direct real-time calculations and the finite-density regime remain difficult. Bottom-up Einstein–Maxwell–dilaton (EMD) models provide a controlled strongly coupled laboratory in which the equilibrium equation of state and retarded response arise from a common black brane geometry.

The critical endpoint is a dynamical as well as an equilibrium problem. Conservation laws and mode coupling determine critical slowing down, and the QCD critical point is expected to belong to Model H [1, 2]. Classical large- N_c holography need not realize the same transport enhancement, so the dynamic exponent and the reorganization of poles

must be measured within the model rather than assumed from the static equation of state [3, 4].

The Bayesian analysis of ref. [5] calibrated two EMD parametrizations to lattice data and inferred a critical endpoint for nearly all posterior samples. Its public posterior preserves the complete model provenance [6], while ref. [7] evaluates thermodynamics and transport across the same phase diagram. This work uses only the maximum-likelihood polynomial-hyperbolic realization. Accordingly, the results below are model predictions at the MAP point, not posterior credible intervals.

Several complementary holographic results delimit the open problem. Homogeneous nonhydrodynamic modes have been followed near critical points in both top-down and QCD-calibrated EMD models [8, 9]. Spinodal sound instabilities and their preferred wavelengths are known in Einstein-scalar models [10–12], and charged-fluid studies show that pole collisions control the convergence of hydrodynamics [13, 14]. Recent improved-holographic-QCD, V-QCD, and Einstein-dilaton calculations further illustrate the variety of finite-density instabilities and channel spectra [15–17]. What remains missing for the PHA realization is a validated treatment of the coupled finite-momentum sectors across thermodynamic branches.

Relative to the homogeneous EMD analysis of ref. [9], the present calculation uses the public PHA maximum-likelihood realization, reconstructs its full equilibrium phase structure, validates every reported pole with a second radial method, and follows all three $SO(3)$ sectors on thermodynamically defined trajectories. It also reproduces nine relaxation times quoted in ref. [9] after substituting that model’s published functions and parameters into the present pipeline. Its novelty is therefore a controlled MAP-model prediction for homogeneous response, not a claim about the dynamic universality class of the endpoint.

Here we give a closed homogeneous-response analysis without assigning coupled-channel physics to decoupled equations. We integrate the exact MAP background in a conserved-charge formulation and compute the quintuplet, triplet, and singlet $SO(3)$ sectors. Source factoring, separate resolution and UV-domain studies, left-right conditioning, eigenfunction overlaps, and a complex shooting calculation validate four poles at the independently located critical background. Three additional eigenvalues of the original discrete pencil fail the shooting test and are discarded.

We separately reproduce the public 2023 constant- ϕ_0 crossing algorithm, locate the cusp from the singular thermodynamic map, and construct the equal-pressure coexistence line. The two local critical determinations agree, but not with the critical coordinates stored in the posterior HDF5 metadata; we report this provenance discrepancy rather than retuning the model. We then continue the validated leading poles along $\mu_B = 0$ and $\mu_B/T = 2$ using shooting and overlap tracking. Coupled finite-momentum dynamics is deliberately outside the claims of this homogeneous paper.

2 PHA EMD model and phase structure

We use natural units, mostly-plus signature, and set the asymptotic AdS radius to one. The action is

$$S = \frac{1}{2\kappa_5^2} \int d^5x \sqrt{-g} \left[R - \frac{1}{2}(\partial\phi)^2 - V(\phi) - \frac{1}{4}f(\phi)F_{MN}F^{MN} \right]. \quad (2.1)$$

The polynomial-hyperbolic Ansatz is

$$\begin{aligned} V(\phi) &= -12 \cosh(\gamma\phi) + b_2\phi^2 + b_4\phi^4 + b_6\phi^6, \\ f(\phi) &= \frac{\operatorname{sech} g(\phi) + d_1 \operatorname{sech}(d_2\phi)}{1 + d_1}, \quad g(\phi) = c_1\phi + c_2\phi^2 + c_3\phi^3. \end{aligned} \quad (2.2)$$

All derivatives used by the solver are analytic. In particular,

$$V_\phi = -12\gamma \sinh(\gamma\phi) + 2b_2\phi + 4b_4\phi^3 + 6b_6\phi^5, \quad (2.3)$$

$$V_{\phi\phi} = -12\gamma^2 \cosh(\gamma\phi) + 2b_2 + 12b_4\phi^2 + 30b_6\phi^4, \quad (2.4)$$

$$\begin{aligned} f_{\phi\phi} &= \frac{1}{1 + d_1} \left\{ \operatorname{sech} g [\tanh^2 g - \operatorname{sech}^2 g] (g')^2 - \operatorname{sech} g \tanh g g'' \right. \\ &\quad \left. + d_1 d_2^2 \operatorname{sech}(d_2\phi) [\tanh^2(d_2\phi) - \operatorname{sech}^2(d_2\phi)] \right\}. \end{aligned} \quad (2.5)$$

Figure 1 displays the resulting functions. The narrow structure in $f(\phi)$ near the boundary is controlled by the large value of d_2 and is numerically resolved rather than smoothed or dropped.

The maximum-likelihood selection gives sample 74 of the public PHA posterior. Table 1 summarizes the realization at a precision suitable for the manuscript; the exact binary64 values and provenance checks are kept in the machine-readable configuration described in appendix A.

The file metadata associates this sample with

$$(T_c, \mu_B^c) = (103.898, 602.475) \text{ MeV}, \quad (2.6)$$

and successful critical-point status. This is provenance metadata rather than the result of the independent phase analysis below.

2.1 Black-brane backgrounds

In domain-wall gauge,

$$ds^2 = e^{2A(r)} [-h(r)dt^2 + d\mathbf{x}^2] + \frac{dr^2}{h(r)}, \quad A_M dx^M = \Phi(r)dt. \quad (2.7)$$

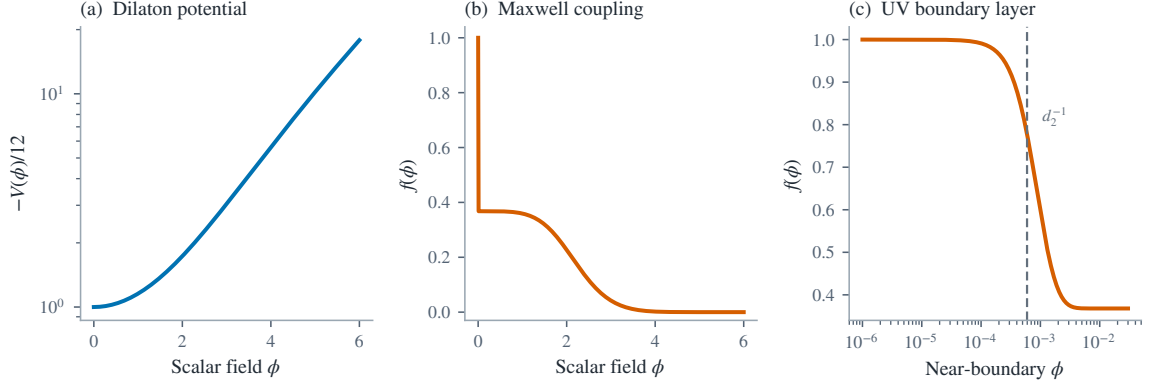


Figure 1. The exact MAP dilaton potential and Maxwell coupling. The right panel resolves the ultraviolet boundary layer at $\phi \sim d_2^{-1}$.

Table 1. Maximum-likelihood PHA realization. Exact machine-readable values and the complete provenance record accompany the numerical data.

quantity	value
Λ [MeV]	1123.87
κ_5^2	11.3705
γ	0.590424
b_2	0.327227
b_4	-0.0480134
b_6	0.000938515
c_1	0.00880048
c_2	0.155799
c_3	0.0528185
d_1	1.71775
d_2	1679.18
m^2	-3.52875
Δ	2.68648

The equations integrated by the reference path are

$$\phi'' + \left(4A' + \frac{h'}{h}\right) \phi' - \frac{1}{h} \left(V_\phi - \frac{1}{2} e^{-2A} f_\phi \Phi'^2\right) = 0, \quad (2.8)$$

$$\Phi'' + \left(2A' + \frac{f_\phi}{f} \phi'\right) \Phi' = 0, \quad (2.9)$$

$$A'' + \frac{1}{6} \phi'^2 = 0, \quad (2.10)$$

$$h'' + 4A'h' - e^{-2A} f \Phi'^2 = 0, \quad (2.11)$$

with constraint and conserved charge

$$\begin{aligned}\mathcal{C} &= h(24A'^2 - \phi'^2) + 6A'h' + 2V + e^{-2A}f\Phi'^2 = 0, \\ \mathcal{Q} &= e^{2A}f\Phi' = \text{constant}.\end{aligned}\tag{2.12}$$

The numerical gauge is $r_H = A_H = h_H = \Phi_H = 0$, $h'_H = 1$, with independent data (ϕ_0, Φ_1) . Regularity requires $|\Phi_1| < [-2V(\phi_0)/f(\phi_0)]^{1/2}$.

The UV coefficients define

$$\begin{aligned}T &= \frac{\Lambda}{4\pi\phi_A^{1/\nu}\sqrt{h_0^{\text{far}}}}, & \mu_B &= \frac{\Phi_0^{\text{far}}\Lambda}{\phi_A^{1/\nu}\sqrt{h_0^{\text{far}}}}, \\ s &= \frac{2\pi\Lambda^3}{\kappa_5^2\phi_A^{3/\nu}}, & \rho_B &= -\frac{\Phi_2^{\text{far}}\Lambda^3}{\kappa_5^2\phi_A^{3/\nu}\sqrt{h_0^{\text{far}}}}.\end{aligned}\tag{2.13}$$

Continuation is carried out in (ϕ_0, Φ_1) , not by isolated inversion in (T, μ_B) . Folds are detected from $\det[\partial(T, \mu_B)/\partial(\phi_0, \Phi_1)] = 0$. Along a line of fixed chemical potential, pressure differences follow from $dP = s dT$; the coexistence temperature is where the integrated pressure difference between the two stable branches vanishes.

Figure 2 shows the resulting first-order region. A 141×121 horizon-data surface supplies 65 constant- μ_B continuations over $594 \leq \mu_B/\text{MeV} \leq 850$. Every line contains two folds and an equal-pressure crossing. The two spinodals and coexistence line meet at the independently located cusp. Repeating the construction on grids coarsened by factors two and four changes the spinodal temperatures by at most 0.018 MeV and the coexistence temperature by at most 0.0085 MeV.

The exploratory scan used below samples 30 logarithmically spaced values $0.18 \leq \phi_0 \leq 5.5$ on each of the slices $\Phi_1/\Phi_1^{\text{max}} = 0, 0.18, 0.32$. Figure 3 shows the direct image of this grid. It is a horizon-data survey, not a phase continuation or Maxwell construction; lines that double back in thermodynamic coordinates must not be read as single-valued equilibrium equations of state.

3 Linear perturbations and QNM sectors

We transform to ingoing Eddington–Finkelstein coordinates,

$$ds^2 = -e^{2A}h dv^2 + 2e^A dv dr + e^{2A}d\mathbf{x}^2, \quad A = \Phi(r)dv,\tag{3.1}$$

and use $e^{-i\omega v + ikz}$. Infalling behavior becomes future-horizon regularity. The primitive-field formulation is chosen so the discretized system is at most linear in frequency,

$$[M_0(k) + \omega M_1(k)]\mathbf{v} = 0.\tag{3.2}$$

This source-free, gauge-invariant prescription follows the standard holographic QNM construction of ref. [18]; coupled-field observables are checked using the operator-mixing framework of ref. [19].

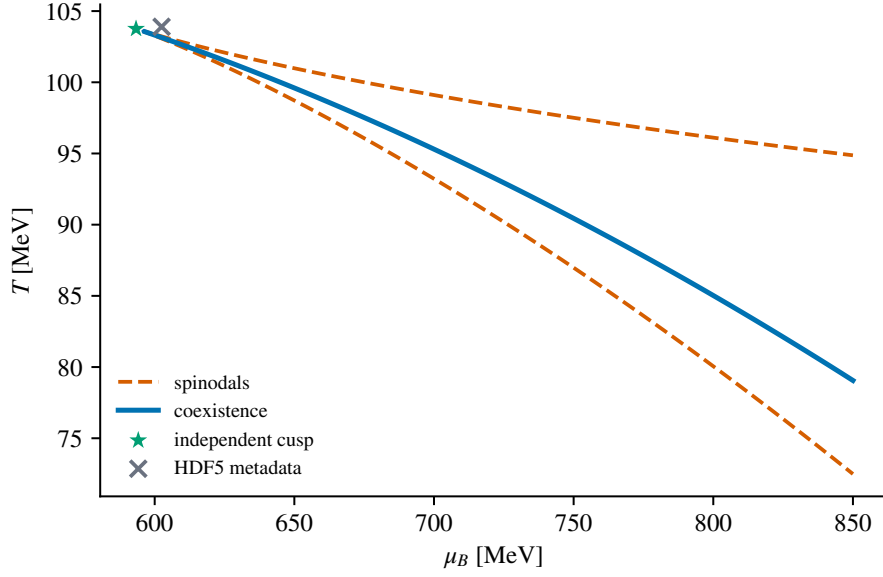


Figure 2. First-order phase structure of the MAP PHA model. Orange dashed curves are the two spinodals, the blue curve is the equal-pressure coexistence line, and the green star is the independent cusp. The gray cross marks the distinct critical coordinates stored in the posterior HDF5 metadata.

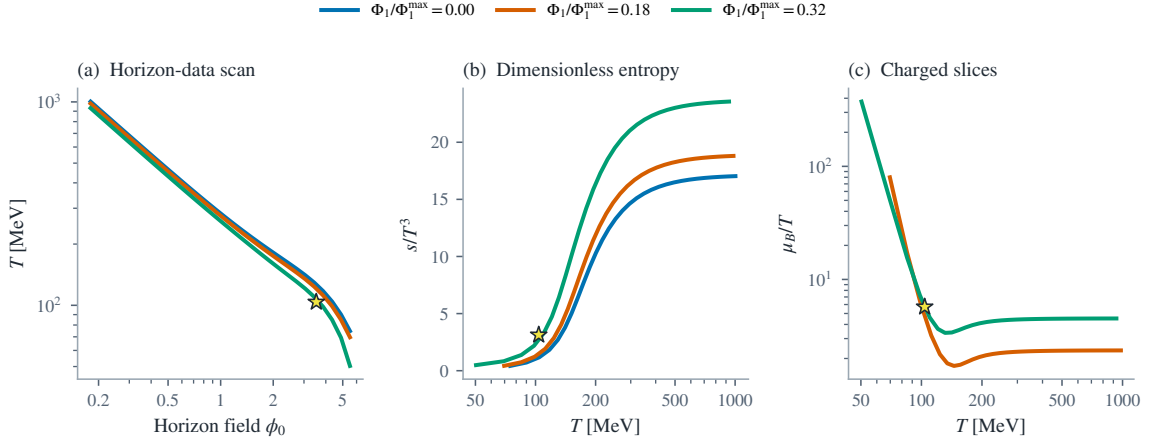


Figure 3. Exploratory background scan in horizon coordinates. The star marks the independently located cusp used for the spectral benchmark. The charged slices expose the nonmonotonic map into $(T, \mu_B/T)$ that makes direct inversion poorly conditioned.

Throughout, ω and k denote physical frequency and momentum, $\mathfrak{w} = \omega/(2\pi T)$ and $\mathfrak{q} = k/(2\pi T)$ their dimensionless counterparts, μ_B the baryon chemical potential, and ρ_B the baryon density. The Fourier convention places stable poles in the lower half of the complex ω plane. Horizon data are denoted by (ϕ_0, Φ_1) ; no symbol in the spectral tables is reused for a thermodynamic branch label.

At $k = 0$, $SO(3)$ organizes perturbations into quintuplet, triplet, and singlet sectors.

At finite k along z , they reorganize into $SO(2)$ helicities two, one, and zero. Radial gauge is used only together with residual-gauge and constraint diagnostics; source conditions are imposed on gauge-invariant combinations.

The tensor/helicity-two equation is

$$Z_2'' + \left(4A' + \frac{h'}{h} - \frac{2i\omega e^{-A}}{h}\right) Z_2' + \left(-\frac{e^{-2A}k^2}{h} - \frac{3i\omega e^{-A}A'}{h}\right) Z_2 = 0. \quad (3.3)$$

At $k = 0$, the homogeneous vector master field obeys

$$a'' + \left(2A' + \frac{h'}{h} + \frac{f_\phi}{f}\phi' - \frac{2i\omega e^{-A}}{h}\right) a' + \left[-\frac{e^{-2A}f\Phi'^2}{h} - \frac{i\omega e^{-A}}{h} \left(A' + \frac{f_\phi}{f}\phi'\right)\right] a = 0. \quad (3.4)$$

The source-free powers are $Z_2 \sim u^4$ and $a \sim u^2$ for $u = e^{-A}$. For the singlet, define

$$\mathcal{S} = \varphi - \frac{\phi'}{2A'} \frac{H_{xx} + H_{yy} + H_{zz}}{3}, \quad Y = \frac{\phi'}{A'}, \quad \psi = \frac{\mathcal{S}}{Y}. \quad (3.5)$$

The homogeneous master equation is

$$\psi'' + \left(4A' + \frac{h'}{h} + 2\frac{Y'}{Y} - \frac{2i\omega e^{-A}}{h}\right) \psi' + \left[-\frac{h'}{h} \frac{Y'}{Y} + \frac{e^{-2A}\Phi'^2}{h\phi'} (3A'f_\phi - f\phi') - \frac{i\omega e^{-A}}{h} \left(3A' + 2\frac{Y'}{Y}\right)\right] \psi = 0. \quad (3.6)$$

An independent first-order variation of the complete EMD system and the field redefinition in eq. (3.5) reproduce eq. (3.6). The same symbolic route generates the coupled finite- k kernels, but their spectra remain behind the numerical gates in appendix D.

4 Numerical method and validation

Two numerical paths are retained. The compiled implementation supplies analytic model kernels, a fourth-order horizon recurrence, background and Chebyshev unit tests, and an independent assembly of the decoupled operators. The reported scans use a separately implemented analysis path. The two routes share the physical conventions but not the background integrator. Software versions and regeneration commands are collected in appendix A.

4.1 Background integration and ultraviolet fit

Rather than evolving both Φ and Φ' , the Python solver fixes $\mathcal{Q} = f(\phi_0)\Phi_1$ at the horizon and reconstructs

$$\Phi'(r) = \frac{\mathcal{Q}e^{-2A(r)}}{f[\phi(r)]}. \quad (4.1)$$

The remaining seven first-order equations are integrated from $\epsilon_H = 10^{-7}$ to $r_{\max} = 12$ with the eighth-order DOP853 method, relative tolerance 3×10^{-12} , absolute tolerance 3×10^{-14} ,

and maximum step 0.025. The ultraviolet coefficients are extracted on a fixed 2400-point sampling grid, so an adaptive solver’s point density cannot bias the fit. The window $10^{-10} < \phi < 10^{-4}$ gives the linear asymptote $A = A_{-1}r + A_0$, while $\Phi_2 = -\mathcal{Q}/(2A_{-1})$ follows directly from Gauss’ law.

Across the 90-point grid the normalized Einstein-constraint residual is at most 3.50×10^{-9} and the reconstructed Gauss-law drift is below 4.45×10^{-16} . The relative spread of ϕ_A within the UV window is between 1.25×10^{-6} and 1.78×10^{-5} . At the background used for the spectra these diagnostics are 1.88×10^{-11} , 3.33×10^{-16} , and 1.35×10^{-6} , respectively. Figure 4 displays its radial profiles.

4.2 Source factoring, conditioning, and independent shooting

The three master equations are collocated on a Chebyshev–Lobatto interval. Future-horizon regularity is built into the ingoing EF equations. In the UV we write $F = e^{-p_F A} y$, with $p_F = 4, 2$, and $2\Delta - 4$ for the quintuplet, triplet, and rescaled singlet fields. The transformed operator is assembled analytically, and the last tau row imposes $y'(r_{\text{cut}}) = 0$. This removes the non-normalizable coefficient at leading asymptotic order without exponentially rescaling the discrete columns. The generalized pencil is passed directly to a QZ driver; no inverse of M_1 is formed [20]. With the numerical horizon normalization used here,

$$\mathfrak{w} = 2\omega_{\text{num}}, \quad k_{\text{num}} = \frac{\mathfrak{q}}{2\sqrt{h_0^{\text{far}}}}. \quad (4.2)$$

Resolution and domain errors are varied separately: $N = 64, \dots, 160$ at $r_{\text{cut}} = 10$, and $r_{\text{cut}} = 8, \dots, 12$ at $N = 160$. Right-eigenfunction overlaps track each branch, while left and right null vectors give $\kappa_\omega = \sqrt{1 + |\omega|^2} \|u\| \|v\| / |u^\dagger M_1 v|$. The largest condition estimate among accepted modes is 2.1×10^5 ; the quoted digits are set by the observed envelopes, not by the algebraic QZ residual.

An independent complex shooting solver starts at $r_0 = 10^{-5}$ with the regular horizon relation, integrates with DOP853, and sets the fitted UV source coefficient to zero. We vary r_0 from 5×10^{-6} to 2×10^{-5} and the extraction radius from 9 to 11. All accepted roots remain within 3.0×10^{-4} in ω_{num} under these changes. Positive-real-part poles are reported once; the reflected partner is implied.

Three roots that passed the original four-grid nearest-frequency gate do not survive shooting: one tensor root converges to a different frequency, while the purely imaginary tensor and triplet roots fail the UV source condition. They are excluded from every result below. This rejection illustrates why a small residual of the discrete pencil is insufficient for a non-normal spectral problem [21, 22].

5 Homogeneous quasinormal spectra

Table 2 lists the four modes that pass both spectral and shooting tests at the independently located critical background. All lie in the lower half plane. The least-damped quintuplet pole is $1.8732(3) - 0.8669(6)i$, whereas the leading triplet lies much closer to the real axis, $1.688542(4) - 0.103168(6)i$. The latter is oscillatory and gapped at $k = 0$; its small damping

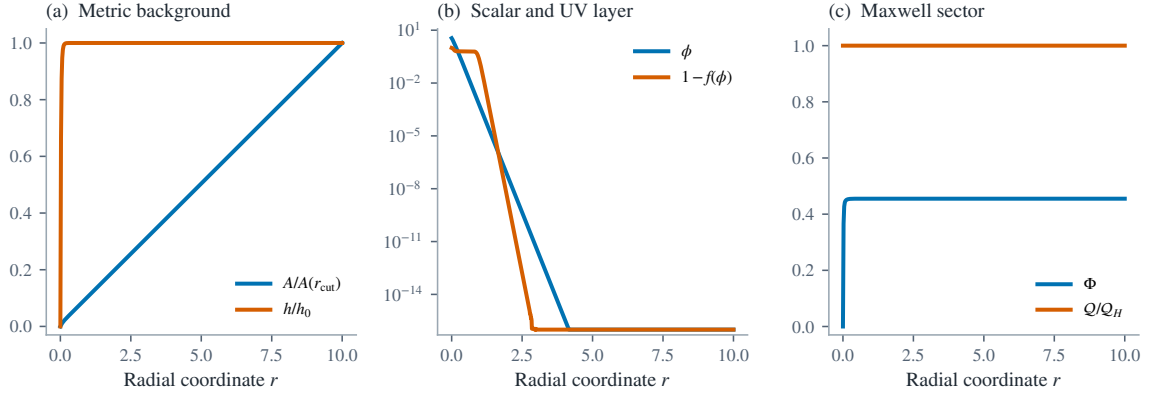


Figure 4. Independently located critical background used for the QNM calculation. The first panel normalizes the metric fields to expose both profiles. The middle panel shows the scalar decay and the narrow Maxwell-coupling boundary layer; the last panel verifies the constant electric flux.

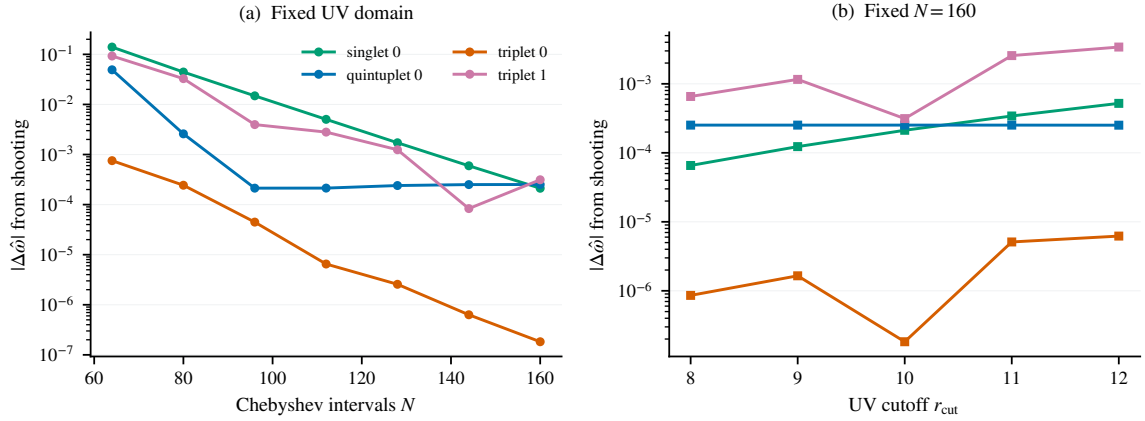


Figure 5. Continuum checks for the four independently validated homogeneous poles. Distances are measured from the complex-shooting root. Resolution and UV-domain variations are separated; the nonmonotonic domain envelope is retained in the reported numerical uncertainty.

is not a baryon-diffusion pole. The leading singlet, $1.4924(12) - 0.2561(13)i$, is longer lived than the quintuplet but shorter lived than the leading triplet.

The relaxation measure associated with a pole is $\tau_{\text{lin}} T = -[2\pi \text{Im } \mathfrak{w}]^{-1}$. It characterizes linear response and is not a far-from-equilibrium thermalization time.

5.1 Controlled thermodynamic trajectories

We continue the leading pole in each sector through 18 backgrounds on each of the curves $\mu_B = 0$ and $\mu_B/T = 2$, spanning approximately $87 \leq T \leq 346$ MeV. Every point is a shooting root, and the corresponding factored-collocation eigenfunction is assigned by frequency and overlap with its parent. The minimum parent overlap is 0.9985; the maximum difference between shooting and the $N = 112$, $r_{\text{cut}} = 8$ spectral root is 0.012 in $\mathfrak{w}$. The complete table also records $(s/T^3, n_B/T^3)$, horizon coordinates, branch and stability labels, distance to the CEP, and the parent index.

Table 2. Independently validated homogeneous poles at $(T, \mu_B) = (103.7553, 593.3227)$ MeV. Parentheses give the conservative spectral/shooting envelope on the last displayed digits.

sector	n	$\mathfrak{w}_n$	source residual	κ_ω
quintuplet	0	$1.8732(3) - 0.8669(6)i$	9.5×10^{-15}	2.1×10^5
triplet	0	$1.688542(4) - 0.103168(6)i$	3.9×10^{-14}	4.6×10^2
triplet	1	$2.8976(34) - 0.8634(12)i$	9.6×10^{-14}	1.4×10^4
singlet	0	$1.4924(12) - 0.2561(13)i$	9.3×10^{-12}	5.1×10^2

Figure 6 shows a common qualitative pattern on the two paths. The quintuplet and singlet damping rates are nonmonotonic but remain of order unity or a few tenths. The leading triplet changes much more strongly: as the low-temperature endpoint is approached it moves to larger real frequency while $-\text{Im } \mathfrak{w}$ falls to 3.7×10^{-3} . A dedicated endpoint study with N up to 160 reproduces this damping rate to 2×10^{-7} in $\mathfrak{w}$, while variations of the shooting start and UV extraction leave the root unchanged within the reported tolerance. This gapped MAP-model triplet is long lived but nonhydrodynamic.

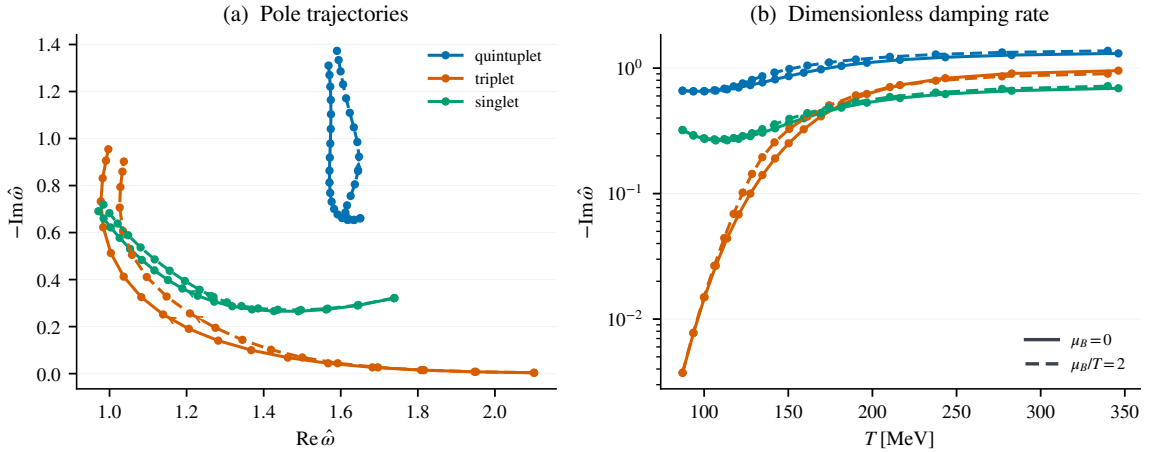


Figure 6. Validated leading homogeneous poles on two controlled thermodynamic trajectories. Arrows in the complex plane point toward higher temperature. Solid and dashed curves denote $\mu_B = 0$ and $\mu_B/T = 2$. The triplet damping decreases by more than two orders of magnitude toward the low-temperature endpoint while its real part stays nonzero.

5.2 Comparison with the 2018 EMD realization

A direct comparison with ref. [9] must distinguish a solver benchmark from model dependence. We therefore replaced the PHA potential, Maxwell coupling and energy scale by the functions and parameters published in that reference, without changing the present collocation or shooting algorithms. A local reconstruction gives the earlier endpoint at (89.122, 723.551) MeV, consistent with the rounded (89, 724) MeV quoted there; the scaled local-cusp residual is 3.1×10^{-4} . At this endpoint and at the two common neutral states used in that paper, all nine published relaxation times are reproduced within 0.0111 fm/c. The largest shooting source residual is 7.5×10^{-9} and the largest spectral-shooting distance is 0.0074 in $\mathbf{w}$.

Table 3 then compares the two models. At $T = 400$ MeV their times differ by at most 0.004 fm/c, displaying the common ultraviolet behavior. At $T = 145$ MeV the PHA singlet relaxes about ten percent faster, while the tensor and triplet remain close. At their respective endpoints both models have the ordering $\tau_3 > \tau_1 > \tau_5$, but the PHA triplet time is about 37 percent larger. Because the endpoint coordinates and calibrated potentials differ, this is a cross-model comparison of homogeneous response, not a critical exponent or a universality statement.

Table 3. Linear relaxation times in the earlier EMD realization and in the PHA MAP model. “Published” values are the approximate plot readings quoted in ref. [9]; “reproduced” values use its parameters in the present solver. Endpoint rows refer to each model’s own CEP.

state	model	T [MeV]	μ_B	τ_5	τ_3 [fm/c]	τ_1
high T	2018 published	400	0	0.060	0.080	0.110
	2018 reproduced	400	0	0.059	0.080	0.108
	PHA MAP	400	0	0.059	0.081	0.112
neutral crossover	2018 published	145	0	0.260	1.000	0.700
	2018 reproduced	145	0	0.261	1.008	0.703
	PHA MAP	145	0	0.261	1.025	0.634
model CEP	2018 published	89.0	724.0	0.330	2.150	1.260
	2018 reproduced	89.1	723.6	0.329	2.139	1.252
	PHA MAP	103.8	593.3	0.349	2.934	1.182

6 Thermodynamic consistency of the finite-density benchmark

We tested the critical point in two independent ways. First, we reconstructed the algorithm distributed with the 2023 model release: second-order horizon data, a relaxation field for the UV coefficient ϕ_A , and crossings of neighboring constant- ϕ_0 trajectories. Increasing the number of trajectories from 100 to 800 moves the crossing estimate from (103.666, 594.761) MeV to (103.761, 593.384) MeV. Second, we solved the local cusp equations. With

$$J = \frac{\partial(\log T, \log \mu_B)}{\partial(\phi_0, \Phi_1/\Phi_1^{\max})}, \quad (6.1)$$

the fold condition is $\det J = 0$ and the cusp additionally requires the directional derivative of $\det J$ along the null vector of J to vanish. This gives

$$(T_c, \mu_B^c) = (103.7553, 593.3227) \text{ MeV}, \quad (\phi_0, \Phi_1/\Phi_1^{\max}) = (3.528281, 0.346374). \quad (6.2)$$

The cusp is within 0.006 MeV in temperature and 0.062 MeV in chemical potential of the 800-trajectory crossing estimate.

Figure 7 displays convergence of the crossing algorithm and the metadata offset. Relative to eq. (6.2), the metadata are higher by 0.143 MeV in temperature and 9.152 MeV in chemical potential, far beyond the stored numerical error estimates. Exact binary64 parameters were used throughout, so rounding was not used to absorb the discrepancy.

The spectral background is the independently located cusp, rather than the nearest inverse image of the metadata. Nevertheless, the weakly damped homogeneous-triplet pole in table 2 is not labeled as critical slowing down: it is gapped, while baryon diffusion and spinodal growth belong to the coupled helicity-zero system.

Table 4. Critical coordinates from the posterior metadata and the two local reproductions. The local methods agree with one another but not with the metadata, which is therefore kept as a provenance value rather than used to select the spectral background.

method	T_c [MeV]	μ_B^c [MeV]
HDF5 metadata	103.8978	602.4749
constant- ϕ_0 crossings ($N = 800$)	103.7611	593.3845
local cusp equations	103.7553	593.3227

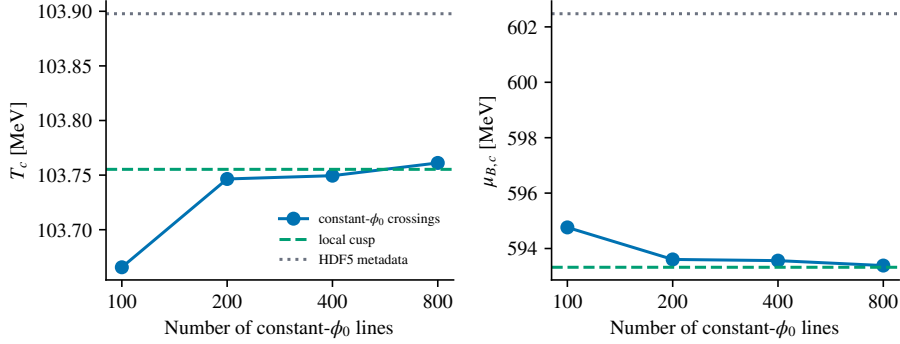


Figure 7. Convergence of the reference-compatible constant- ϕ_0 crossing algorithm. The independent local cusp agrees with the converged trajectory estimate; the HDF5 metadata do not.

7 Conclusions

For the maximum-likelihood PHA realization, source-factored collocation and independent shooting validate the leading homogeneous quintuplet and singlet poles and two triplet poles at the finite-density critical background. The least-damped values are $\mathbf{w}_5 = 1.8732(3) - 0.8669(6)i$, $\mathbf{w}_3 = 1.688542(4) - 0.103168(6)i$, and $\mathbf{w}_1 = 1.4924(12) - 0.2561(13)i$. Three apparent roots of the original discrete pencil are rejected by the independent source condition.

The same implementation reproduces the nine relaxation times quoted for the earlier EMD model in ref. [9]. This cross-model benchmark recovers their common high-temperature limit and shows a quantitative change in the infrared: at the respective endpoints the PHA triplet relaxation time is about 37 percent longer, whereas both models retain the ordering $\tau_3 > \tau_1 > \tau_5$.

Along $\mu_B = 0$ and $\mu_B/T = 2$, shooting and overlap tracking give the same qualitative ordering. The most pronounced feature is the gapped triplet pole: its dimensionless damping falls by more than two orders of magnitude and reaches 3.7×10^{-3} at the low-temperature endpoint. The pole remains oscillatory, and the endpoint value is stable under both spectral refinement and shooting-domain variations. The quintuplet and singlet show milder, nonmonotonic damping minima.

The equilibrium calculation now independently fixes the state. It resolves the two

spinodals and equal-pressure coexistence line, and two different cusp algorithms agree. Their result differs from the HDF5 critical metadata by 9.15 MeV in chemical potential; the discrepancy is recorded without changing the exact MAP parameters. Conserved-charge evolution controls Maxwell drift, while the neutral entropy and susceptibility are stable under the tested finite-difference steps.

The scope is intentionally homogeneous. Coupled finite-momentum helicity-one and helicity-zero calculations are required to establish shear, sound, baryon diffusion, critical slowing, or spinodal growth, and none is inferred here. Likewise, posterior uncertainty must be propagated before these MAP results can be described as Bayesian predictions for real-time observables.

The interpretation retains the limitations of large N_c , strong coupling, translational invariance, equilibrium backgrounds, and the bottom-up PHA construction. The calculation concerns QNMs of a lattice-calibrated holographic EMD model, not QNMs of QCD.

Acknowledgments

OpenAI Codex was used as an AI-assisted tool during preparation of this work for code refactoring, workflow automation, construction of numerical and symbolic validation scripts, literature-metadata checks, and language and layout editing. Its outputs were checked against the version-controlled source, machine-readable results, cited primary literature, and independent numerical tests. Responsibility for the scientific content and the submitted manuscript remains with the authors.

A Reproducibility and data products

The reported run used Python 3.12.13, NumPy 2.5.2, SciPy 1.18.1, and Matplotlib 3.11.1. The source HDF5 checksum matches the value in the data manifest; scanning the posterior groups selects `sample74`, with $\log \mathcal{L} = 31.0751$. The full checksum and exact binary64 metadata remain in the machine-readable configuration. From the repository root, the complete analysis is regenerated by

```
python -m pip install -r analysis/requirements.txt
python analysis/reproduce_reference_cep.py --lines 800 --neighbors-only \
    --output results/python/reference_cep_neighbors_N800.json
python analysis/locate_reference_cusp.py --phi-step 0.01 \
    --charge-step 0.001 --uv-tolerance 1e-5 \
    --output results/python/reference_cusp_uv1e5_h10.json
python analysis/build_phase_diagram.py --n-phi 141 --n-charge 121 \
    --phi-min 2.2 --phi-max 6.5 --charge-min 0.02 --charge-max 0.62 \
    --mu-min 594 --mu-max 850 --mu-count 65
python analysis/validate_phase_diagram.py
python analysis/plot_cep_reproduction.py
python analysis/run_numerics.py
python derivations/generate_linearized_emd.py
```

```
python derivations/validate_linearized_emd.py
python analysis/validate_decoupled_qnms.py
python analysis/build_homogeneous_trajectories.py
python analysis/compare_2018_emd.py
python analysis/plot_qnm_validation.py
python analysis/plot_homogeneous_trajectories.py
```

The numerical-results directory contains the thermodynamic surface, constant- μ_B branch table, grid-convergence tables, background and susceptibility scans, independently validated homogeneous modes, and both controlled thermodynamic trajectories. The cross-model table stores both the published plot readings and the independently reproduced 2018-model values. JSON sidecars record the exact parameters, numerical settings, UV coefficients, branch construction, cusp residuals, background diagnostics, QNM residuals, and resolution and shooting envelopes. Every manuscript figure is emitted in PDF and PNG form under `paper/figures/`. The versioned source code and machine-readable outputs supporting this article are publicly available at <https://github.com/RenatoPhys/pha-qnm>; the upstream posterior data are archived with ref. [6].

The dense generalized eigenproblem is deterministic for a fixed LAPACK build; minor last-digit changes are possible across QZ implementations. The quoted precision is therefore deliberately below the machine residual and is set by the observed grid/cutoff displacement. No random sampling enters this analysis.

B Neutral thermodynamics and baryon susceptibility

As a numerical cross-check independent of the QNM pencil, we reproduce the neutral entropy density and the second-order baryon susceptibility

$$\chi_2^B(T) = \left. \frac{\partial(\rho_B/T^3)}{\partial(\mu_B/T)} \right|_{\mu_B=0}. \quad (\text{B.1})$$

For each of the 30 neutral backgrounds, the derivative is evaluated with the centered ratio

$$\chi_2^B(\delta) = \frac{(\rho_B/T^3)_{+\delta} - (\rho_B/T^3)_{-\delta}}{(\mu_B/T)_{+\delta} - (\mu_B/T)_{-\delta}}, \quad \delta = \frac{\Phi_1}{\Phi_1^{\max}}, \quad (\text{B.2})$$

at $\delta = 0.005, 0.010$, and 0.020 . Charge-conjugation symmetry makes T and s even and μ_B and ρ_B odd in δ ; consequently, fixing ϕ_0 in this centered derivative differs from fixing T only at $\mathcal{O}(\delta^2)$.

Figure 8 reports all 90 derivative evaluations. On the finest step, the scan covers $74.69 \leq T/\text{MeV} \leq 1003.70$, $0.418 \leq s/T^3 \leq 17.033$, and $1.66 \times 10^{-6} \leq \chi_2^B \leq 0.31869$. The largest displacement relative to the $\delta = 0.005$ result is 2.06×10^{-3} , attained by the coarsest step. Across the same runs, the normalized Einstein-constraint residual remains below 3.50×10^{-9} and Gauss-law drift below 4.45×10^{-16} . The complete values and diagnostics are stored in `results/python/thermodynamic_validation.csv`; no smoothing or fitted interpolation enters the displayed points.

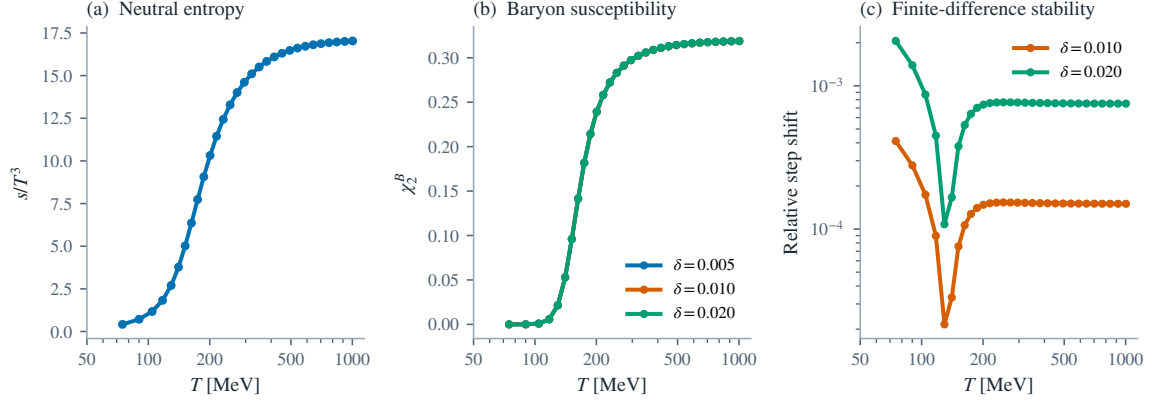


Figure 8. Reproduction and numerical stability of neutral thermodynamics. Left: dimensionless entropy. Center: χ_2^B from the centered charge derivative in eq. (B.2); the three step sizes are visually indistinguishable on this scale. Right: relative displacement from the finest-step result. This is a $\mu_B = 0$ equilibrium validation and is not a finite-density CEP or spinodal claim.

C Regular horizon expansion

Writing

$$\begin{aligned} A &= A_1 r + A_2 r^2 + \dots, & h &= r + h_2 r^2 + \dots, \\ \phi &= \phi_0 + \phi_1 r + \phi_2 r^2 + \dots, & \Phi &= \Phi_1 r + \Phi_2 r^2 + \dots, \end{aligned} \quad (\text{C.1})$$

the compiled recurrence reproduces

$$\begin{aligned} A_1 &= -\frac{2V_0 + f_0 \Phi_1^2}{6}, & \phi_1 &= V_{\phi,0} - \frac{1}{2} f_{\phi,0} \Phi_1^2, \\ A_2 &= -\frac{\phi_1^2}{12}, & h_2 &= \frac{1}{2} (f_0 \Phi_1^2 - 4A_1), \\ \Phi_2 &= -\frac{\Phi_1}{2} \left(2A_1 + \frac{f_{\phi,0}}{f_0} \phi_1 \right). \end{aligned} \quad (\text{C.2})$$

Higher coefficients through fourth order are generated by truncated-series arithmetic applied to the regular equations, rather than hand-entered formulas.

D Validation and claim scope

The authoritative machine-readable tables store background horizon data, physical thermodynamics, branch and stability labels, frequency, resolution, pencil and source residuals, condition number, tracking parent, and eigenfunction overlap. A mode enters a figure or table only when continuum convergence and independent shooting both pass. Generated finite-momentum equations and their symbolic identity tests are supplied as reproducibility artifacts, but no coupled finite-momentum spectrum enters the scientific claims of this paper.

References

- [1] P.C. Hohenberg and B.I. Halperin, *Theory of dynamic critical phenomena*, *Rev. Mod. Phys.* **49** (1977) 435.
- [2] D.T. Son and M.A. Stephanov, *Dynamic universality class of the QCD critical point*, *Phys. Rev. D* **70** (2004) 056001 [[hep-ph/0401052](#)].
- [3] O. DeWolfe, S.S. Gubser and C. Rosen, *Dynamic critical phenomena at a holographic critical point*, *Phys. Rev. D* **84** (2011) 126014 [[1108.2029](#)].
- [4] J.V. Roth, Y. Ye, S. Schlichting and L. von Smekal, *Universal critical dynamics near the chiral phase transition and the QCD critical point*, *Phys. Rev. D* **111** (2025) L111901 [[2409.14470](#)].
- [5] M. Hippert, J. Grefa, T.A. Manning, J. Noronha, J. Noronha-Hostler, I. Portillo Vazquez et al., *Bayesian location of the qcd critical point from a holographic perspective*, *Phys. Rev. D* **110** (2024) 094006 [[2309.00579](#)].
- [6] M. Hippert, J. Grefa, T.A. Manning, J. Noronha, J. Noronha-Hostler, I. Portillo Vazquez et al., *Bayesian analysis of the equation of state of quantum chromodynamics from a holographic model*, 2024. 10.5281/zenodo.13830379.
- [7] M.R. Khan, A. Nascimento, Y. Yang, J. Grefa, M. Hippert, J. Noronha et al., *Uncertainty quantification of holographic transport and energy loss for the hot and baryon-dense qgp*, [2603.20482](#).
- [8] S.I. Finazzo, R. Rougemont, M. Zaniboni, R. Critelli and J. Noronha, *Critical behavior of non-hydrodynamic quasinormal modes in a strongly coupled plasma*, *JHEP* **01** (2017) 137 [[1610.01519](#)].
- [9] R. Rougemont, R. Critelli and J. Noronha, *Nonhydrodynamic quasinormal modes and equilibration of a baryon dense holographic qgp with a critical point*, *Phys. Rev. D* **98** (2018) 034028 [[1804.00189](#)].
- [10] R.A. Janik, J. Jankowski and H. Soltanpanahi, *Nonequilibrium dynamics and phase transitions in holographic models*, *Phys. Rev. Lett.* **117** (2016) 091603 [[1512.06871](#)].
- [11] R.A. Janik, J. Jankowski and H. Soltanpanahi, *Quasinormal modes and the phase structure of strongly coupled matter*, *JHEP* **06** (2016) 047 [[1603.05950](#)].
- [12] A. Caddeo, O. Henriksson, C. Hoyos and M. Sanchez-Garitaonandia, *Spinodal slowing down and scaling in a holographic model*, *JHEP* **08** (2024) 091 [[2406.15297](#)].
- [13] N. Abbasi and S. Tahery, *Complexified quasinormal modes and the pole-skipping in a holographic system at finite chemical potential*, *JHEP* **10** (2020) 076 [[2007.10024](#)].
- [14] A. Jansen and C. Pantelidou, *Quasinormal modes in charged fluids at complex momentum*, *JHEP* **10** (2020) 121 [[2007.14418](#)].
- [15] T. Alho, J. Remes, K. Tuominen and A. Vuorinen, *Quasinormal modes and thermalization in improved holographic QCD*, *Phys. Rev. D* **101** (2020) 106025 [[2002.09544](#)].
- [16] J. Cruz Rojas, T. Demircik and M. Jarvinen, *Modulated instabilities and the AdS_2 point in dense holographic matter*, *Phys. Rev. D* **111** (2025) 046017 [[2405.02399](#)].
- [17] N.A. Villarreal, L.A.H. Mamani, A. Ballon-Bayona, A.S. Miranda and V.T. Zanchin, *Transport coefficients and quasinormal modes in einstein-dilaton holographic QCD*, *Phys. Rev. D* **112** (2025) 066017 [[2507.04165](#)].

- [18] P.K. Kovtun and A.O. Starinets, *Quasinormal modes and holography*, *Phys. Rev. D* **72** (2005) 086009 [[hep-th/0506184](#)].
- [19] M. Kaminski, K. Landsteiner, J. Mas, J.P. Shock and J. Tarrio, *Holographic operator mixing and quasinormal modes on the brane*, *JHEP* **02** (2010) 021 [[0911.3610](#)].
- [20] A. Jansen, *Overdamped modes in schwarzschild-de sitter and a mathematica package for the numerical computation of quasinormal modes*, *Eur. Phys. J. Plus* **132** (2017) 546 [[1709.09178](#)].
- [21] J.L. Jaramillo, R.P. Macedo and L. Al Sheikh, *Pseudospectrum and black hole quasinormal mode (in)stability*, *Phys. Rev. X* **11** (2021) 031003 [[2004.06434](#)].
- [22] K. Destounis, R.P. Macedo, E. Berti, V. Cardoso and J.L. Jaramillo, *Pseudospectrum of reissner–nordstrom black holes: quasinormal mode instability and universality*, *Phys. Rev. D* **104** (2021) 084091 [[2107.09673](#)].